

In Search of Cost-Reflective Electricity Tariffs and Cost Allocation Principles: A Guide for African Regulators, Policy Makers and Utilities

INTRODUCTION

Across Africa, electricity regulators face the momentous challenge of ensuring reliable, secure, and affordable power, while also safeguarding the financial sustainability of power utilities. At the heart of this complex challenge lies the need for cost-reflective tariffs—pricing structures that not only accurately reflect the true costs of electricity generation, transmission, and distribution, but also protect consumers and promote economic growth.

This succinct guide provides regulators and utilities with practical, high-level insights into implementing cost-reflective tariffs, focusing on proven cost-allocation principles, methodologies, and best practices that can be adapted to African contexts.

UNDERSTANDING THE REGULATORY CHALLENGE

The Case for Tariff Regulation

In most African nations, electricity remains a natural monopoly, with one or a few utilities controlling the market. Without effective incentive regulation, these utilities can exploit their market power, potentially charging excessive rates and/or providing substandard service. Conversely, political pressure can push tariffs below cost-recovery levels, undermining utilities' ability to maintain and expand their infrastructure. Either scenario compromises the electricity supply industry that underlies economic development and national stability.

Effective incentive and tariff regulation serves multiple critical functions: it ensures utilities remain financially viable while protecting consumers from excessive monopolistic prices, promotes efficient resource allocation, encourages efficient infrastructure investment, and maintains service quality standards. However, achieving these objectives requires careful balance and robust methodological frameworks.

Core Regulatory Objectives

Successful tariff regulation must consider and balance four fundamental objectives:

- 1) **Economic and financial sustainability:** utilities must be able to cover their economic costs. This includes a fair and reasonable risk-adjusted return on investments, and sufficient cashflows to maintain operations for continuity of supply.

- 2) **Allocative efficiency:** prices must be set to direct resources to their most valuable use; that is, they must reflect the true cost of service and provide proper signals for decision making so that consumption is aligned with efficient production and equitable distribution and use.
- 3) **Productive efficiency:** the ability to provide electricity at the lowest cost (minimizing service delivery costs).
- 4) **Equity and affordability:** ensuring fair access for all, which includes pricing in line with the ability to pay for poor consumer groups.

These objectives often are in tension, requiring regulators to make careful trade-offs. For instance, the drive for cost recovery might conflict with affordability goals, while the pursuit of efficiency might temporarily compromise equity objectives. Equally, incumbent monopoly utilities' desire to set prices that reflect the cost of service disincentivizes efforts to reduce those costs, as such cutbacks would result in lower tariffs and more constrained cashflows. When firms or utilities generate excess/ free cashflows (beyond what is necessary for profitable investments), managers may be tempted to invest it in projects and activities that do not have a positive net present value leading to inefficiencies.

Regulators therefore must balance these objectives while also responding to constantly shifting internal and external factors. **This balancing act is the basis of regulation.** Regulators, therefore, must be afforded sufficient information to prioritize, verify,

and implement these objectives; that is, the data that reveals the industry's intrinsic characteristics (e.g., degree of competition; structure of industry, the amount and type of investments needed; the economic, social, and political environment, etc.).

THE FOUNDATION: COST ALLOCATION

Understanding Costs

Any effective tariff system begins with a clear understanding of costs. **Direct costs**—those readily traceable to specific services or activities—form the simplest component.

The challenge lies in dealing with **indirect costs**, which fall into two categories:

- **Joint Costs:** These arise when a single process produces multiple outputs simultaneously. Not easily separable, joint costs need to be allocated appropriately across different services provided by the utility. For example:
 - A power plant might generate both electricity and heat, making it difficult to separate the costs between these products.
 - Investments in grid improvements (such as smart meters or renewable energy integration technologies) carry costs that are considered joint if they benefit both generation and distribution.
- **Common Costs:** These are shared across services that could otherwise be produced separately.
 - Administrative overhead represents a classic example, as it supports multiple services.

Goals for cost allocation

Cost allocation can have multiple goals, not all of which align. Overall, cost allocation should support at least some of the following:

- accurate price determination and transparency
- evaluation of performance and profitability
- cost reimbursement
- incentives to use specific services within a utility more or less intensively.

Regulators need to prioritize which of these goals to pursue, and make their rationale for their choice clear to the public and government.

Allocation Principles

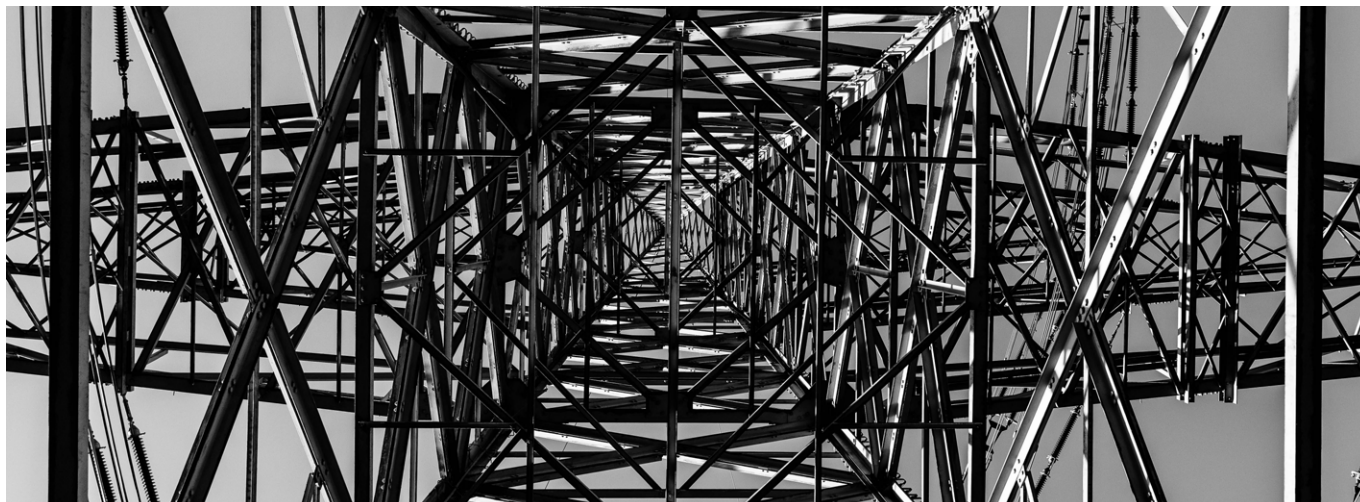
Regulators can approach cost allocation through four criteria:

- **Causality principle:** costs are assigned to the activities that incur them, providing the most economically efficient basis for allocation.
- **Benefits-received:** costs are distributed in proportion to the benefits different groups derive from those services.
- **Ability-to-bear:** the different users' financial capacity is considered in allocating costs.
- **Arbitrary allocation:** used as a last resort when the other approaches prove impractical.

Cost allocation and cost exclusion

For clarity, in terms of regulatory accounting, the allocation of costs is understood to mean the allocation of **prudent and efficient costs**, NOT just any cost that is incurred. Regulators have a right not only to audit relevant data, but also to exclude¹ some costs in full (unregulated costs), or to only partially approve incurred expenses considered to be inefficient² or not useful.

In other words, even when correctly allocated, not all costs should necessarily be used for tariff determination. From an economic perspective, only expenses *useful* for delivering the regulated service should be recovered through tariff charges to the users of the regulated service (for example, inefficient costs should not be recovered from customers through tariffs).



METHODOLOGICAL APPROACHES

Cost allocation methodologies have evolved significantly over time, each offering distinct advantages and challenges.

Fully Distributed Costs (FDC): The simplest and most straightforward approach, FDC uses broad, imprecise cost drivers (e.g., units sold, revenue generated, etc.) to allocate all costs.

- **Pros:** Simple and straightforward.
- **Cons:** Lacking economic efficiency, FDC's arbitrary nature (based on a simple ratio) can hide inefficiencies, create unintended cross-subsidies, and encourage incentives for cost padding.

Activity-Based Costing (ABC): A more sophisticated approach than FDC, ABC links costs to specific activities and processes (actions, work sequences, events, transactions, or time in-puts) rather than final goods or services outputs. That is, ABC first links costs to processes, then links those processes back to final goods or services outputs.

- **Pros:** Greater accuracy and transparency, particularly in allocating indirect costs.
- **Cons:** Requires detailed operational knowledge and sophisticated data collection systems—resources that many African regulators may find challenging to access or maintain. This complexity perpetuates the information asymmetry gap, and can create tensions between the regulator and utility.

Hybrid Cost Allocations (HCA): HCA uses integrated technological platforms to automate and allocate costs using recently developed billing

systems linked to financial accounting systems (e.g., Enterprise Resource Planning or ERP; SAP Ariba software).

- **Pros:** Provides a balance of benefits from the FDC and ABC options. Requires adopting a Uniform System of Accounts (USoA)³ to harmonize asset and service definitions for transparent cost coding and allocation between utilities and regulators
- **Cons:** Like ABC, requires detailed systematic data collection tools and systems. The challenge of this is that most utilities are financially distressed unable to invest in such modern technology platforms and often, input data is difficult to find as it is usually manually stored.

Ramsey Pricing: An economic approach based on demand elasticity, Ramsey allocates costs based on different consumer groups' or services' sensitivity to price changes. That is, a larger share of costs are allocated to activities or products that are less sensitive to price variations.

- **Pros:** Theoretically elegant and economically efficient
- **Cons:** Requires extensive data on demand patterns and elasticities, making it challenging to implement in practice.

Regardless, with the advent of Artificial Intelligence (AI), all the above methods are increasingly becoming possible to implement. Utilities and regulators need to invest in research and development to take advantage of these technologies.

IMPLEMENTATION FRAMEWORK

The Path to Cost-Reflective Tariffs

Implementing cost-reflective tariffs requires a systematic approach that begins with careful activity identification during cost of service studies. Regulators must work with utilities to map core **operational activities** and determine relevant **cost drivers**⁴. This foundation enables accurate cost assignment, where direct costs are allocated to specific activities, and indirect costs are grouped into appropriate **cost pools**⁵.

The development of tariff components follows, linking costs to specific elements of the pricing structure while considering factors such as time-of-use and customer categories. Throughout this process, performance monitoring provides essential feedback on the effectiveness of cost allocation and the impact of tariffs on both utilities and consumers.

Fundamental Implementation Challenges

Two fundamental challenges often impede effective implementation:

- **Information asymmetry:** regulators cannot effectively function as oversight bodies of pricing methodologies without utilities' cooperation in sharing detailed information about their operations and costs. Regulators must be able to exercise their legal mandate and tools (revoking licenses, imposing penalties – pre-agreed prior to imposition) to compel utilities to share information if needed.
- **Technical capacity constraints:** limited data, analytical capabilities and resource shortages which restrict regulators' ability to implement more sophisticated methodologies.

Cost distortions and cross-subsidies

When costs are improperly assigned to cost objectives, cross-subsidies and distortions occur. The activity to which some of its own costs were not assigned will be subsidized by the activities to which these costs were improperly assigned. This subsidization will lead regulators and utilities to make wrong decisions for the given activities on the basis of (distorted) profitability analyses and price determination studies. It is therefore important that regulators and utilities properly understand this.

Cross-subsidization may arise when utilities:

- fail to trace costs directly to the industry segments when the situation requires it
- use inappropriate cost driver bases to allocate costs
- arbitrarily allocate common costs to industry segments

RECOMMENDATIONS FOR SUCCESS

Building Regulatory Capacity

Success in implementing cost-reflective tariffs requires sustained investment in regulatory capacity building. Regulators should prioritize:

- Developing robust information systems and standardized reporting requirements
- Building technical expertise in cost allocation and tariff design
- Establishing clear, transparent processes for tariff review and adjustment, including unbundling of tariffs

Practical Implementation Steps

Implementation should follow a phased approach, starting with simpler methodologies and gradually increasing complexity as capacity grows. Regular monitoring, adjustment, and rigorous evaluation of the system's impacts (on electricity prices, utilities' financial sustainability, and broader economic and social outcomes) are necessary to ensure the system remains effective and responsive to changing conditions.

- Define, clearly articulate, and correct:
 - Gaps in cost allocation approaches; specifically around cost allocation objectives & indirect costs.
 - Inconsistencies around asset valuation approaches
- Practical pathways for regulators to verify compliance with cost objectives are needed. These may include:
 - Investment in robust data collection mechanisms (e.g., smart meters, automated systems)
 - Longitudinal (5-years) studies on ability and willingness to pay to enable the build-up of baselines for sensitivity analyses.
- Consistency in implementation of the pricing rule: whatever methodology is adopted, consistent application of the rules without cherry-picked ad-hoc modifications is key to a stable and investor-confident regulatory environment.

Consumer Protection

While pursuing cost-reflectivity, regulators must not lose sight of consumer protection. This includes designing progressive tariff structures that shield vulnerable consumers while maintaining utility sustainability. Targeted subsidies, clearly separated from tariff structures, can help achieve social objectives without distorting price signals.

CONCLUSION

Implementing cost-reflective tariffs or revenue-sufficient systems represents a crucial step in realizing sustainable electricity sectors across Africa. Regulators and regulated companies both operate in an environment rife with external uncertainties and pressure to meet challenging targets with limited information. As such, regulators must be able to proactively assess how incentive and tariff regulation needs to evolve in order to respond to the rapid energy transition, changing power markets, and other

external changes (e.g., the costs of new technologies or government policy regarding the scaling up or down of certain technologies). Careful attention to methodological rigor, appropriate cost allocation, practical implementation challenges, and stakeholder interests are all critical. By following the principles and approaches outlined in this guide, regulators can work toward pricing systems that support both utility sustainability and broader economic development goals.

Endnotes

- 1 Regulators want to avoid the risk that the incurred expenses are charged at a rate higher than their market value – incentive regime.
- 2 When a regulator decides to exclude inefficient costs, regulated utilities will want to know the legal grounds for this action, and this must be clearly indicated in law, contract, or regulation.
- 3 USoA is a standardized method of classifying, recording, and reporting financial and operational information for electricity utilities.
- 4 Cost-drivers can be categorized as input-based, output-based, or revenue-based (e.g., machine hours, labour hours, proportion of resource used)
- 5 Used to allocate overhead and other indirect costs, cost pools aggregate indirect costs that are shared across multiple cost objects (e.g., products or departments)

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